

Glucoraphanin and 4-hydroxyglucobrassicin contents in seeds of 59 cultivars of broccoli, raab, kohlrabi, radish, cauliflower, brussels sprouts, kal...

The importance of dietary sulforaphane in helping maintain good health continues to gain support within the health-care community and awareness among U.S. consumers. In addition to the traditional avenue for obtaining sulforaphane, namely, the consumption of appropriate cruciferous vegetables, other consumer products containing added glucoraphanin, the natural precursor to sulforaphane, are now appearing in the United States. Crucifer seeds are a likely source for obtaining glucoraphanin, owing to a higher concentration of glucoraphanin and the relative ease of processing seeds as compared to vegetative parts. Seeds of several commonly consumed crucifers were analyzed not only for glucoraphanin but also for components that might have negative health implications, such as certain indole-containing glucosinolates and erucic acid-containing lipids. Glucoraphanin, 4-hydroxyglucobrassicin, other glucosinolates, and lipid erucic acid were quantified in seeds of 33 commercially available cultivars of broccoli, 4 cultivars each of kohlrabi, radish, cauliflower, Brussels sprouts, kale, and cabbage, and 2 cultivars of raab.